$\text{Max } Z = 40x_1 + 60x_2$

Avec $x_1 ; x_2 \geq 0$.

s.c : $8x_1 + 5x_2 \leq 40$

$6x_1 + 9x_2 \geq 54$

$x_1 \geq 2$

Forme Standard :

$\text{Min } Z = -40x_1 - 60x_2$

s.c : $\begin{cases} 8x_1 + 5x_2 + e_1 = 40 \\ 6x_1 + 9x_2 - U_1 = 54 \\ x_1 - U_2 = 2 \end{cases}$

$(x_1 ; x_2 ; U_1, U_2, e_1 \geq 0)$



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To

	x_1	x_2	e_1	u_1	u_2	C
	8	5	1	0	0	40
X	6	9	0	-1	0	54
	1	0	0	0	-1	2

$\neq I_3$.

Forme standard :

$$\text{Min } Z = -40x_1 - 60x_2$$

Sc

$$\begin{cases} 8x_1 + 5x_2 + e_1 = 40 \\ 6x_1 + 9x_2 - u_1 + w_1 = 54 \\ x_1 - u_2 + w_2 = 2 \end{cases}$$

$(x_1 ; x_2 ; e_1 ; u_1 ; u_2 ; w_1 ; w_2 \geq 0)$

$x_1 = 0$
$x_2 = 0$
$Z = 0$

To'

		x_1	x_2	e_1	u_1	u_2	w_1	w_2	C
(BS)	$0\ e_1$	8	5	1	0	0	0	0	40
	$0\ w_1$	6	9	0	-1	0	1	0	54
	$0\ w_2$	1	0	0	0	-1	0	1	2
	Z_A	40	60	0	0	0	0	0	0

2

$Z = Z_j - C_j$

Phase I : Changement d'objectif :

$\text{Min } Z_A = \omega_1 + \omega_2 \quad (\omega_1 + \omega_2 = 0)$

$+ \begin{cases} \omega_1 = 54 - 6x_1 - 9x_2 + u_1 \\ \omega_2 = 2 - x_1 + u_2 \end{cases}$

$\text{Min } Z_A = 56 - 7x_1 - 9x_2 + u_1 + u_2$

$$\text{Sc} : \begin{cases} 8x_1 + 5x_2 + e_1 = 40 \\ 6x_1 + 9x_2 - u_1 + \omega_1 = 54 \\ x_1 - u_2 + \omega_2 = 2. \end{cases}$$

$$\begin{cases} x_1 = 0 \\ x_2 = 0 \\ u_1 = 0 \\ u_2 = 0 \end{cases}$$

$Z_A = 56$

Avec $(x_1 ; x_2 ; e_1 ; u_1 ; u_2 ; \omega_1 ; \omega_2 \geq 0)$

Tableau 1

		x_1	x_2 (VE)	e_1	u_1	u_2	ω_1	ω_2	C	K
0	e_1	8	5	1	0	0	0	0	40	8
0	ω_1	6	9	0	-1	0	1	0	54	6
0	ω_2	1	0	0	0	-1	0	1	2	✗
	Z_A	7	9	0	-1	-1	0	0	56	

(VP : ω_1, ω_2)

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T'_{02}

	x_1 (VB)	x_2	e_1	u_1	u_2	w_1	w_2	C	K
e_1	14/3	0	1	5/9	0	-5/9	0	10	15/7
x_2	6/9	1*	0	-1/9	0	1/9	0	6	9
(VP) w_2	(1)	0	0	0	-1	0	1	2	2
ZA	1	0	0	0	-1	-1	0	(2)	

1C	8	5	1	0	0	0	0	40	
1P	6/9	1*	0	-1/9	0	1/9	0	6	x-5 +
1R	14/3	0	1	5/9	0	-5/9	0	10	

1C	7	9	0	-1	-1	0	0	56	
1P	6/9	1*	0	-1/9	0	1/9	0	6	x-9 +
1R	1	0	0	0	-1	-1	0	2	

T'_{03}

	x_1	x_2	e_1	u_1	u_2	w_1	w_2	C
e_1								
x_2								
x_1	1	0	0	0	-1	0	1	2
ZA								

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LC	14/3	0	1	5/9	0	-5/9	0	10
LP	1*	0	0	0	-1	0	1	2
LR	0	0	1	5/9	14/3	-5/9	-14/3	2/3

$\times -\frac{14}{3}$ +

LC	6/9	1	0	-1/9	0	1/9	0	6
LP	1*	0	0	0	-1	0	1	2
LR	0	1	0	-1/9	6/9	1/9	-6/9	14/3.

$\times -\frac{6}{9}$ +

LC	1	0	0	0	-1	-1	0	2
LP	1*	0	0	0	-1	0	1	2
LR	0	0	0	0	0	-1	-1	0 //

$\times -1$ +

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T'_{02}

	x_1 (Ve)	x_2	e_1	u_1	u_2	ω_1	ω_2	C	K
	14/3	0	1	5/9	0	-5/9	0	10	15/7
e_1	6/9	1*	0	-1/9	0	1/9	0	6	9
x_2	6/9	1	0	0	0	-1	0	1	2
(Vp) ω_2	(1)	0	0	0	-1	0	1	2	
	1	0	0	0	-1	-1	0	(2)	

Z_A

LC	8	5	1	0	0	0	0	40	
LP	6/9	1*	0	-1/9	0	1/9	0	6	x-5 +
LR	14/3	0	1	5/9	0	-5/9	0	10	

LC	7	9	0	-1	-1	0	0	56	
LP	6/9	1*	0	-1/9	0	1/9	0	6	x-9 +
LR	1	0	0	0	-1	-1	0	2	

T'_{03}

	x_1	x_2	e_1	u_1	u_2	ω_1	ω_2	C
e_1	0	0	1	5/9	14/3	-5/9	-14/3 + 2/3	
x_2	0	1	0	-1/9	6/9	1/9	-6/9 + 14/3	
x_1	1	0	0	0	-1	0	1	2
Z_A	0	0	0	0	0	-1	-1	0

→ Light afficht.

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Phase II : Retour à l'objectif initial : $\min$

$$\min Z = -40x_1 - 60x_2$$

T_1		x_1	x_2	e_1	y_1 (VE)	y_2	C	K
0	e_1 (VD)	0	0	1	5/9	14/3	2/3	6/5
-60	x_2	0	1	0	-1/9	6/9	14/3	Négatif
-40	x_1	1	0	0	0	-1	2	α
	Z	0	0	0	20/3	0	-360	

$\Downarrow$ $f = (2)$.

$Z_j - C_j$

$$Z = (-40 \times 2) - (60 \times \frac{14}{3}) = -360$$

X X

	x_1	x_2	e_1	y_1	y_2	C
y_1	0	0	9/5	1*	42/5	6/5
x_2	0	1	1/5	0	8/5	24/5
x_1	1	0	0	0	-1	2
Z	0	0	-12	0	-56	-368

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LC	0	1	0	-1/9	6/9	14/3
LP	0	0	9/5	1*	42/5	6/5
LR	0	1	1/5	0	8/5	24/5

)x 1/9] +

LC	0	0	0	20/3	0	-360
LP	0	0	9/5	1*	42/5	6/5
LR	0	0	-12	0	-56	-368

) x -20/3] +



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